Cognisphere v3.4

Formal Mathematical Specification
Personal Cognitive Architecture Model

Individual Cognitive Model (Single Subject)

March 2026

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1 Formal Definition

$$CS_{v3.4} = \langle \mathcal{S}_{space},\, \mathcal{L}_{layers},\, \mathcal{D}_{dynamics},\, \mathcal{F}_{functions},\, \mathcal{M}_{meta},\, \mathcal{I}_{inputs} \rangle$$

$$\begin{aligned}
\mathcal{S}_{space} &= \{S(t),\, R(t),\, x\} \\
\mathcal{L}_{layers} &= \{L_0,\, L_1,\, L_1',\, L_2,\, M,\, K\} \\
\mathcal{D}_{dynamics} &= \{t,\, W,\, D,\, V,\, P,\, T,\, F,\, \Psi,\, r,\, \Xi\} \\
\mathcal{F}_{functions} &= \{G,\, \text{Depth},\, \text{Rot},\, \zeta,\, \Lambda_{scale},\, L,\, \Phi\} \\
\mathcal{M}_{meta} &= \{\Phi_{L_0},\, SD,\, R_{recovery},\, \kappa\} \\
\mathcal{I}_{inputs} &= \{q,\, \Theta\}
\end{aligned}$$

Note: $\Xi(t) = (T(t),\, F(t),\, P(x,t),\, K,\, r(t))$ is a bundled expression of $\mathcal{D}_{dynamics}$.

2 Space Definition

$$S(t) \subset \mathbb{R}^4 \quad \text{(master space, 4D including time axis)}$$

$$\begin{aligned}
&S_0(t) \subseteq S(t) \quad \text{Layer 0 domain, nested structure } M(t) \in [1,6] \\
&S_1(t) \subseteq S(t) \quad \text{Layer 1 domain (invariant, minor deformation at limit)} \\
&S_1'(t) = G(\Theta, t, S(t)) \subseteq S(t) \quad \text{Layer 1' domain (auto-deforms with } \Theta, t)
\end{aligned}$$

3 Layer 0: Basal Constant System

$$\Phi_{L_0} \equiv \text{Const} \quad (\forall\, t,\, \forall\, r)$$

$$\begin{aligned}
&\text{Activation} \perp r(t) \quad \text{(Layer 0 activation independent of } r(t)) \\
&\text{Observation density, load} \sim \text{sleep state, pathology, physical condition}
\end{aligned}$$

Self-inhibition maximization (upon overload):

$$SD(t) \nearrow 1 \implies t_{limit} \text{ executed} \quad \text{(volitional stop, not forced termination)}$$

4 K-Flag Definition

$$K \in \{0, 1\}, \quad K \text{ discrete}$$

$$\begin{aligned}
K = 0 &\implies \text{operation rules unknown} \implies L_1 \text{ activated} \\
K = 1 &\implies \text{operation rules known} \implies L_1' \text{ activated}
\end{aligned}$$

Note: Determined by the known/unknown status of operation rules, not the target itself.

K-judgment process:

$$q \xrightarrow{\text{K-judgment}} \begin{cases} L_1 & K = 0 \\ L_1' & K = 1 \end{cases}$$

In most cases, initial scaffolding forms from known concepts (K=1 equivalent). When operation rules exceed known range:

$$K : 1 \to 0 \implies L_1 \text{ takes over}$$

5 Layer Transition Rules

Exclusive simultaneity + sequential transitionability:

$$L_1 \cap L_1' = \emptyset \quad \text{(no simultaneous parallel operation)}$$

$$\exists\, t^* : \; K(t^*) \text{ transitions} \implies \text{dominant Layer transitions at } t^*$$

Dominant Layer at any time t is unique (one-to-one).

6 Plasticity Parameter $r(t)$

$$r(t) \in [0, 1]$$

Definition: Structural flexibility / change-deformation tolerance of cognitive architecture.

$$r \approx 0 \implies \text{rigid: strict convergence conditions, limits, tolerance range}$$
$$r \approx 1 \implies \text{plastic: soft convergence, large tolerance range}$$

Derived quantities (first-order derivatives of r):

- Softness of convergence judgment
- Increase in sustained thinking capacity
- Ease of model update/evolution
- Relaxation of upper limit conditions in L_1/L_1'

6.1 Change Law

$$\frac{dr}{dt} = \eta_{\text{eff}}(t) - \lambda(\text{age}, \text{style}) \cdot r(t) - \delta_{rest}, \quad \delta_{rest} \geq 0$$

Effective ascent rate (η hierarchy):

$$\eta_{\text{base}} \in \{\eta_{normal},\, \eta_{PEM},\, \eta_{introspect},\, \eta_{combined}\}$$

$$\eta_{\text{eff}}(t) = \eta_{\text{base}} \cdot e^{-\kappa t}$$

where κ is the growth decay coefficient (distinct from $\gamma(r)$ convergence sensitivity).
State-wise base rates:

$$\begin{aligned}
\eta_{normal} &: \text{standard AI feedback loop} \\
\eta_{PEM} &: \text{PEM crash state} \quad (\eta_{normal} \gg \eta_{PEM} > 0) \\
\eta_{introspect} &\approx 0.5\, \eta_{normal} \quad \text{(introspection only)} \\
\eta_{combined} &= \eta_{normal} \times (1 + \beta \cdot D_{internal}(t))
\end{aligned}$$

Quadratic growth ($\eta_{combined}$):

$$D_{internal}(t) = \alpha t, \quad f_{combined}(t) = \eta_{normal}(1 + \beta\alpha t) \cdot t \quad \text{(contains } t^2 \text{ term)}$$

Simulation-confirmed values ($r_0 = 0.30$, weekly scale):

$$\begin{aligned}
\frac{r(18\,\text{wk})}{r_0} &\approx 2.98 \quad \text{(introspection + AI, actual baseline)} \\
\frac{r(18\,\text{wk})}{r_0} &\approx 2.78 \quad \text{(AI loop only, theoretical)} \\
\frac{r(18\,\text{wk})}{r_0} &\approx 1.80 \quad \text{(severe PEM crash, always } r > 0.05\text{)}
\end{aligned}$$

7 Termination Conditions

Layer 1:

$$\text{Terminate} \iff T(t) \geq T_{max}(r) \ \vee \ \Phi(\Theta, t, r) \approx 0 \ \vee \ t \geq t_{limit}(r)$$

Layer 1':

$$\text{Terminate} \iff F(t) \geq F_{max}(r) \ \vee \ \Phi(\Theta, t, r) \approx 0 \ \vee \ t \geq t_{limit}(r)$$

Common (both layers):

$$W(0^+) \approx W_{max} \quad \text{(instant max load, no ramp-up cycle)}$$

Note: T_{max} = subjective thermal limit; t_{limit} = volitional external constraint (PEM avoidance).

8 Stop Decision Function $SD(t)$

$$SD(t) = \text{Sigmoid}\left(\int_t^{t+H_{pred}} C_{future}(PEM, \tau)\, d\tau - Threshold_{will}\right)$$

$$\begin{aligned} C_{future}(PEM, \tau) &: \text{predicted future life-impairment cost} \\ H_{pred} &: \text{future prediction horizon} \\ Threshold_{will} &: \text{volitional stop threshold (varies with } r(t)) \end{aligned}$$

$SD(t) \to 1$: t_{limit} executed. Continuous decision process (Sigmoid, not Step).

9 Convergence Evaluation Function Φ

$$\Phi(\Theta, t, r) = w_1(\Theta, r)\, \Phi_{prac} + w_2(\Theta, r)\, \Phi_{abst} - \gamma(r) \cdot \text{Stability}(P, V)$$

$$\text{Stability}(P, V) = \frac{1}{1 + a\|\nabla P\| + b\|\partial V/\partial t\|}$$

Convergence condition by domain Θ:

$$\Phi_0(\Theta) = \begin{cases} \text{generality} \wedge \text{practicality} \wedge \text{individual applicability} & \Theta \in \mathcal{D}_{prac} \\ \text{logical exhaustion} & \Theta \in \mathcal{D}_{abst} \end{cases}$$

Convergence $\iff$ $\Phi(\Theta, t, r) \approx 0$ ("quiet landing", not pressure release).

10 Thinking Process (Layer 1)

$$q \xrightarrow{K=0} W(0^+) \to W_{max}$$

$$\text{Depth}(h_i) = \int_{S_{h_i}} D(x, t)\, dx, \quad h^* = \arg\max_i \text{Depth}(h_i)$$

$$\text{Depth}(h^*)\uparrow \implies h_{new} \text{ generated (generative chain)}$$

Convergence condition:

$$\nabla P(x, t) \to 0 \ \wedge \ \frac{\partial V}{\partial t} \to 0 \quad (\text{Stability} \to 1)$$

11 Spatial Operations (Layer 1')

$$\begin{aligned} \text{Zoom in/out} : x' &= \zeta x, \quad \zeta < 1 \text{ (in)}, \ \zeta > 1 \text{ (out)} \\ \text{Viewpoint rotation} : x' &= \mathrm{Rot}(\theta)\, x \\ \text{Scale switch} : \Lambda_{scale} &: S \to S' \quad \text{(discontinuous, discrete)} \end{aligned}$$

12 Verbalization Operator $L(\Psi, r)$

$$\text{Output} = L(\Psi, r)$$

$$\Psi \in \begin{cases} \text{4D image } (\mathrm{L}_0) \\ \text{3D high-density fluid convergence state } (\mathrm{L}_1) \\ \text{variable space convergence state } (\mathrm{L}_1') \end{cases}$$

L: high-dimensional concept field $\to$ high-density compressed structured text.

13 Fatigue Recovery Dynamics $R_{recovery}$

$$\text{PEM_flag} \in \{0, 1\} \quad \text{(PEM state flag)}$$

Normal recovery:

$$\left.\frac{dF}{dt}\right|_{recovery} = -k_{normal}\, F(t)$$

PEM state (PEM_flag = 1)**:**

$$\left.\frac{dF}{dt}\right|_{PEM} = -k_{PEM}\, F(t), \quad k_{PEM} \ll k_{normal}$$

13.1 Origin: Handwritten Sketch and Final Figure Correspondence

This PEM pseudo-recovery pattern was first defined as a phenomenon structure based on internal observation (handwritten sketch), then mathematically formalized. The final figure is not an AI-generated artifact, but the result of formalizing the time evolution, ceiling structure, and reactivation delay contained in the sketch.

Seven structures defined in the sketch:

1. Peak = 1.0 ceiling: saturation-type upper limit
2. 0.2 residual load baseline: not complete recovery
3. Saturating rise during work: approaches ceiling asymptotically
4. Rapid drop under parasympathetic dominance: phase-transition-like
5. Low-value maintenance as pseudo-recovery: internally fragile
6. 20–30 min reactivation delay after restart: delayed rebound
7. Re-adhesion to Peak: activity strongly re-drives the ceiling

13.2 PEM Pseudo-Recovery Pattern (Simulation Confirmed)

Local symbols:

$$\begin{aligned} F_0 &: \text{fatigue at reactivation start} \\ t_{work} &: \text{assumed session work time upper limit (e.g., 60 min)} \\ F_{peak_prev} &: \text{fatigue peak reached in previous session} \end{aligned}$$

Fatigue rise during work (simulation-confirmed):

$$F_{work}(t) \approx F_0 + (1.0 - F_0) \cdot \frac{\ln(1 + t/s)}{\ln(1 + t_{work}/s)}$$

Checkpoints: ∼0.6 @ 20 min, ∼0.8 @ 40 min, 1.0 @ 60 min ($s \approx 60$).

$$F(t) \leq 1.0 \quad \text{(ceiling: working memory full load state)}$$

Apparent recovery (parasympathetic dominant):

$$F_{apparent}(t) = F_{true}(t) \cdot (1 - \varepsilon_{para}), \quad 0 < \varepsilon_{para} < 1$$

Post-reactivation overshoot:

$$F(t + \Delta t) > F_{peak_prev} \quad (\Delta t \lesssim 20\,\text{min, typically less})$$

Reactivation: rapid ascent to exceed F_{peak_prev}, then gradual approach to 1.0.
Complete recovery is undefined in this model. Chronic fatigue accumulation is the baseline state.

14 Layer Relationship

$$L_0 \xrightarrow{\text{constant supply}} q \xrightarrow{K\text{-judgment}} \begin{cases} L_1 & (K = 0) \\ L_1' & (K = 1) \end{cases} \xrightarrow{\Phi \approx 0} L_2$$

$$L_1 \cap L_1' = \emptyset, \quad \text{sequential transition: } L_1 \rightleftharpoons L_1' \text{ allowed}$$

$$W(0^+) \approx W_{max} \text{ for both } L_1, L_1'$$

15 Cognitive Load Profile

Layer	W	Duration	T rise	F accum.	Stop/Suppression trigger
L_0	low–mid	constant	none	chronic	no stop in principle / suppressed by sleeping pills
L_1	W_{max}	2–5h	large (occipital)	yes	$T_{max}(r)$ / $\Phi \approx 0$ / $t_{limit}(r)$
L_1'	W_{max}	3–6h	small	large	$F_{max}(r)$ / $\Phi \approx 0$ / $t_{limit}(r)$
L_2	high, brief	short	small	small	verbalization complete

16 Pathological Constraint Conditions

Note: Based on hypothetical organization through iterative dialogue with multiple LLMs, not clinical diagnosis.

Condition	Primary Layer	Numerical Effect
PEM	L_1, L_1'	η_{PEM} decrease, $k_{PEM} \ll k_{normal}$, t_{limit} tightened
SAD	L_0	observation density maximized, load surge
C-PTSD	L_0	hypervigilance, observation cost chronic
Attachment disorder	L_0	boundary fluctuation in romantic context
Sleep disorder	all	$F(t)$ chronic accumulation, recovery failure
Bipolar type 2	$L_1, L_1', r(t)$	manic: W excess / depressive: $\eta_{combined}$ facilitated